Study notes — for teammates new to this topic

Read time ~25 minutes. Goal: after reading, you can follow every slide of `trackb.pdf`, answer the obvious questions, and know which numbers are memorable. Nothing here requires prior quantum-information background beyond "a qubit is a two-level system and circuits apply gates to it."

Companion documents: `EXPLAINER.md` (gentler, non-specialist), `RESULTS.md` (every number's source row), `deck/SPEAKER_SCRIPT.md` §Q&A (rehearsed answers).

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0. The one-paragraph version

The Hadamard test is a standard circuit that measures one helper qubit (the *ancilla*) to extract a number χ(t) from a quantum system — and throws the rest of the qubits away as "garbage." A 2025 paper (Faehrmann, Eisert & Kueng, PRL 135, 150603) pointed out you can measure that garbage in random bases and get extra information for free. Our project implements that idea, extends it (Track B: compare *two* dynamics in one circuit), makes the expensive part cheap with a circuit compiler (AQC), finds and fixes a trap that silently breaks the whole scheme, and then tests everything on real quantum computers — including the test that *failed*, which we report with the same prominence as the wins.

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1. The Hadamard test (slide: "You compiled U…", appendix A1–A2)

What it computes. For a state |ψ⟩ and evolution U, the complex number χ = ⟨ψ|U|ψ⟩. Its magnitude says "how much does U leave ψ alone"; its **phase** carries the dynamics (energies live in phases: U = e^{−iHt}).

How. Put the ancilla in a superposition (Hadamard gate), apply U to the system *only if* the ancilla is |1⟩ (controlled-U), interfere (second Hadamard), measure the ancilla. Then ⟨Z_ancilla⟩ = Re χ. Repeat with a small phase gate (φ = −π/2) to get Im χ.

The one thing to internalise: the Hadamard test is an *interferometer*. Anything blind to phase — a magnitude, a fidelity — is blind to what this circuit measures. That single sentence explains three separate findings below.

Memorise: χ is complex; |χ| ≤ 1 always. If a measurement reports |χ| > 1, the experiment is broken, full stop (this happens on slide "We ran it on a QPU" — deliberately shown).

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2. Classical shadows (slide: "First, what a classical shadow is", A4)

Problem it solves. Measuring a qubit destroys it, and one measurement basis gives one kind of information. If you fix a basis in advance, you can only ever estimate observables diagonal in that basis.

The trick. Each shot, pick each qubit's basis (X, Y or Z) *at random*, and store just `(bases, outcomes)`. A weight-w Pauli observable (one acting non-trivially on w qubits) matches the drawn bases with probability 3^−w; multiply matching shots by 3^w and average. The estimator is *unbiased for every Pauli at once* — you did not have to choose what to measure in advance.

The price. Variance = 3^w per shot. Low-weight observables are cheap; weight-n observables are exponentially expensive. Our project measured this model to be exact within 2% [R029].

In this project: the garbage register of the Hadamard test is measured this way. Tagging each shadow shot with the ancilla outcome (±1) links the system information to the interference signal.

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3. Track B: the anti-controlled Hadamard test (slides "Anti-control…",

"Track B…", "What the shadows add"; A1–A3, A5)

The question it answers. You want to run U but can only afford an approximation W (a Trotterised circuit, a compiled circuit). How wrong is W — and wrong *in what*?

The circuit change. Sandwich the controlled-W between two X gates on the ancilla. X flips $|0\rangle \leftrightarrow |1\rangle$, so the sandwich makes W fire on the $|0\rangle$ branch while the ordinary controlled-U fires on the $|1\rangle$ branch. The ancilla now interferes *two dynamics*: $\langle Z_a \rangle_\varphi = \mathrm{Re}[e^{i\varphi} \langle\psi|W^\dagger U|\psi\rangle]$. Setting W = identity gives back the standard test — Track B strictly generalises it.

Our completion (the part the scaffold left open). Delete the *final* Hadamard. Then, of the shadow data: the plain average estimates $(W\rho W^\dagger + U\rho U^\dagger)/2$ — the SUM — and the ancilla-weighted average estimates $(W\rho W^\dagger - U\rho U^\dagger)/2$ — the DIFFERENCE. Add and subtract: you get every observable under W and under U *separately*, from one circuit family. Validated to 4×10^{-15} [R042].

Slogan (use it in Q&A): *the ancilla says how much W is wrong; the garbage says which observable it got wrong.*

Honest context [R043, R044, R059]: if you only want the scalar "how close is W to U", the *Loschmidt echo* (run U forward, W backward, measure the probability of returning to start — ~2 gates here) is 4–6× cheaper and, on hardware, 27–32× more accurate than our arm (widening over a 2×-extended time window, R059). The *Hilbert–Schmidt test* answers a state-independent version on 2n qubits. Our arm earns its keep only through the phase and the per-observable profile. The deck says this out loud — do not soften it, it is our credibility.

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4. AQC and the crossover (slide "AQC makes it scale"; R046, R049, R051, R052)

The bottleneck. The textbook way to build controlled-U for an arbitrary U synthesises a (n+1)-qubit unitary: cost $\sim 4^{(n+1)}$ two-qubit gates. Doubling from n=3 to n=6 takes you from 50 to 4,140 gates (8,850 after routing to a real chip).

AQC (Approximate Quantum Compiling). A classical optimiser (tensor-network based) finds a *shallow* circuit whose action on your state matches the deep one. Its cost grows roughly linearly in n. Hence a crossover: exact wins below n=4, AQC wins above — 36× fewer gates by n=7, at state infidelity $\sim 3\times10^{-4}$. The advantage survives 2D lattices, the Fermi–Hubbard model (crossover shifts to 6 qubits), and real heavy-hex routing — routing actually *widens* the gap, because the dense exact block routes worse than the shallow local ansatz.

One scope caveat to remember: AQC compresses the action on *one input state* (process infidelity ~0.96!). Fine for a Hadamard test — the controlled gate only ever sees that state. Illegal anywhere else.

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5. The phase trap — the intellectual heart of the deck (slide "The trap";

R046, R053, R054)

AQC maximises **state fidelity** $|\langle\psi_{target}|\psi_{ansatz}\rangle|$ — a *magnitude*. The Hadamard test measures a *phase* (§1). So a compression can converge perfectly by its own metric and return $W|\psi\rangle = e^{i\theta}U|\psi\rangle$ with θ up to ~3 radians: χ comes out wrong by ~1.3 on a quantity bounded by 1. **The error is as large as the signal, and no fidelity number warns you.**

The fix: θ is computable for free during compilation; apply P(−θ) — one single-qubit phase gate — on the ancilla. Error drops 1.33 → 0.008.

Why we call it a pattern, not an incident — three independent sightings: 1. The compiler's objective (above) [R046]. 2. A summary statistic: "survival" = |χ_measured|/|χ_exact| looked near-perfect (0.988) on a teammate's `ibm_miami` run whose *complex* χ was 22.6σ wrong [R053]. 3. Our own hardware run: the deadest circuit (8,850 gates, ancilla fully relaxed) got the *best* survival score, 1.485 — unphysical, since |χ| ≤ 1 [R054].

Q&A soundbite: *a Hadamard test is an interferometer; any magnitude-only figure of merit is structurally blind to what it measures.*

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6. The hardware reality check (slides "We ran it on a QPU", "First Track B

on a QPU", "free error bar"; R044, R054, R055)

Design. Same Hadamard test at n=6, three ways to build the controlled evolution (exact 8,850 routed gates / Trotter 2,119 / AQC 576), one job per device, survival predictions written down *before* submission — so failure would be undeniable, and it was: every arm on both `ibm_marrakesh` and `ibm_kingston` returned noise; the AQC arm hit 0.026/0.033 against predicted 0.315/0.401 — 12× short on both.

The post-mortem number to remember: effective error per two-qubit gate, bounded from data: $\varepsilon_{eff} \geq -\ln(S)/N \approx 6\times10^{-3}$ (a lower bound — the measured |χ| sits near the estimator's noise floor ~0.020 at these shots) — about 3–4× the *calibrated* gate error. Ancilla dephasing (T_2) explains part; a ~7× residual is honestly labelled unattributed. The win becomes visible when $\varepsilon_{eff} \lesssim 2\times10^{-3}$ — roughly one device generation away (Nighthawk's target regime). Full arithmetic: appendix A7.

Why n=6 and not n=7: the deliberately-dead exact *control* arm dominates cost (602 µs/shot; ~4× more at n=7) and the contrast was already saturated. "Doubles the headline, halves the signal, quadruples the bill." [R055]

The one clean hardware win: ⟨Q⟩_W − ⟨Q⟩_U ≡ 0 exactly (charge conservation survives Trotterisation), so its measured deviation is pure device error — an error bar with no reference state, no simulation. Stays bounded (not growing) over a 2×-extended time window, t up to 5.4. [R044, R059]

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7. Numbers worth memorising (all sourced)

number	what it is	row
36×	AQC gate reduction at n=7 (compilation)	R046
n=4	AQC/exact crossover	R046
1.33 → 0.008	χ error before/after the one-gate phase fix	R046
0.026 / 0.033 vs 0.315 / 0.401	measured vs pre-registered AQC survival, two QPUs	R054
1.485	the unphysical "survival" of a dead circuit	R054
$\sim6\times10^{-3}$ vs $\lesssim2\times10^{-3}$	effective error today vs what visibility needs	R054/A7
27× (32× extended)	how much the 2-gate echo beats our arm on hardware	R044, R059
4×10^{-15}	Track B identity validation	R042
12/12	seeds correct on the Track A deliverable	R019
56/56	notebook checkpoints passing	— (README)

8. Glossary (30 seconds each)

- **ancilla** — the helper qubit whose measurement carries the interference.
- **χ (chi)** — ⟨ψ|U|ψ⟩; complex; |χ| ≤ 1.
- **quadrature** — the φ=0 / φ=−π/2 runs giving Re χ and Im χ.
- **Pauli weight w** — number of qubits an observable touches; shadows pay 3^w.
- **Trotterisation** — approximating e^{−iHt} by products of small steps.
- **routing / transpiling** — mapping a circuit onto a chip's actual layout; inserts SWAPs, increases gate count.
- **T_1 / T_2** — decay times: energy relaxation / phase coherence. Circuit time beyond these ⇒ the qubit forgets.
- **survival** — |χ_measured|/|χ_exact|. Useful, but magnitude-only: treat any survival quoted *without* the complex χ with suspicion (that is our own lesson, thrice).
- **pre-registered** — prediction recorded before the experiment ran, so the outcome can falsify it.
- **effective error ε_eff** — −ln(survival)/gate-count: the per-gate error the device *behaved* as having, all noise sources folded in.

9. If a judge asks you and you don't know

Say: "That's in our results ledger — every number on the slides has a row tag, and the row lists the command that produced it. I can pull it up." That answer is always true, always safe, and demonstrates the project's core discipline. The rehearsed answers for the hard questions (n=6 vs n=7, ratio vs absolute, the priority claim, "did you try SKQD?") are in `SPEAKER_SCRIPT.md` §Q&A.